

CHE 311 HW 3, Chapter 3

(1)

Question 2: By making a mass balance over a volume element $(\Delta r)(r\Delta\theta)(\Delta z)$ derive the equation of continuity in cylindrical coords.

Eq: $\left[\frac{\partial \rho}{\partial t}\right] + (\nabla \cdot \rho \mathbf{V}) = 0$



APP. B: $\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$
(B.4-2)

$(\Delta r)(r\Delta\theta)(\Delta z) \frac{\partial \rho}{\partial t} = \overset{\text{surface 1}}{(\rho v_r|_{r=0} - \rho v_r|_{r=\Delta r}) r \Delta\theta \Delta z} + \overset{\text{surface 2}}{(\rho v_\theta|_{\theta=0} - \rho v_\theta|_{\theta=\Delta\theta}) \Delta r \Delta z} + \overset{\text{surface 3}}{(\rho v_z|_{z=0} - \rho v_z|_{z=\Delta z}) \Delta r r \Delta\theta}$

$\frac{\partial \rho}{\partial t} = \frac{(\rho v_r|_{r=0} - \rho v_r|_{r=\Delta r})}{\Delta r \cdot r} + \frac{(\rho v_\theta|_{\theta=0} - \rho v_\theta|_{\theta=\Delta\theta})}{\Delta\theta \cdot r} + \frac{(\rho v_z|_{z=0} - \rho v_z|_{z=\Delta z})}{\Delta z \cdot r}$

$+ \frac{(\rho v_r|_{r=0} - \rho v_r|_{r=\Delta r})}{\Delta r \cdot r} + \frac{(\rho v_\theta|_{\theta=0} - \rho v_\theta|_{\theta=\Delta\theta})}{\Delta\theta \cdot r} + \frac{(\rho v_z|_{z=0} - \rho v_z|_{z=\Delta z})}{\Delta z \cdot r}$

$\frac{\partial \rho}{\partial t} = - \left(\frac{\partial \rho v_r}{\partial r} + \frac{1}{r} \frac{\partial \rho v_\theta}{\partial \theta} + \frac{\partial \rho v_z}{\partial z} \right)$

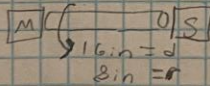
$\frac{\partial \rho}{\partial t} = - \left(\frac{\partial \rho v_r}{\partial r} + \frac{1}{r} \frac{\partial \rho v_\theta}{\partial \theta} + \frac{\partial \rho v_z}{\partial z} \right)$

or find $(\nabla \cdot \mathbf{V})$ in APP. A7 in cylindrical

3.A.2: Answer: 0.115

$v_\theta = v_\theta(r)$

- Shaft rotates at $500 \text{ rpm} = \Omega$
- Lubricant = $\mu = 500 \text{ cP}$
- 20 bearings, 1 ft each in length



• Sleeve bearings that give .005 in clearance

• Use cylindrical

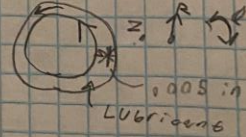
Continuity (Const M, ρ)

$0 = 0 + 0 + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) = 0$

$v_\theta|_{r=\sin} = R(\sin)$

$r(\text{in})$

$v_\theta|_{r=8.005 \sin} = 0$



Motion App. 2

(2)

$$r: \rho\left(-\frac{v\phi^2}{r}\right) = -\frac{2\rho}{2r} + \mu(0+0+0+0)$$

$$\phi: \rho(0+0+\frac{v\phi}{r}+\frac{v\phi}{r}+0+0)$$

$$= -\frac{1}{r}(\rho) + \mu\left[\frac{2}{2r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv\phi)\right)+0+0+0\right]$$

$$z: \rho(0+0+0+0) = -\frac{2\rho}{2z} + \mu(0+0+0+0)$$

$$\phi: \rho = \frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rv\phi)\right)$$

$$rc_1 = \frac{\partial}{\partial r}(rv\phi) \rightarrow \left\{ \frac{c_2}{r} + \frac{rc_1}{2} = v\phi \right\}$$

$$BC: 0 = \frac{8.005c_1}{2} + \frac{c_2}{8.005} \rightarrow c_2 = -\frac{8.005^2 c_1}{2}$$

$$r(8in) = \frac{8c_1}{2} + \frac{c_2}{8}$$

$$\rightarrow r(8) = \frac{8c_1}{2} + \left(\frac{8.005^2 c_1}{8}\right) \rightarrow r(8) = 4c_1 + \frac{8.005^2 c_1}{8}$$

$$r(8) = c_1(4 - 8.005^2) \rightarrow c_1 = -r(1600)(17)$$

$$r(8) = \frac{8c_1}{2} + \frac{c_2}{8} \quad c_2 = \frac{8.005^2}{8}(-1600)$$

$$v\phi = -\frac{r(1600)}{2} - \frac{8.005^2(-1600r)}{2}$$

$$v\phi = 12in = \frac{r}{2} \left(\frac{1600}{2} - \frac{8.005^2}{2} \right) \quad \frac{r(5000)}{2} = \frac{r(5000)}{2}$$

CH 8.21
HW3 (3)

$$\frac{\partial \rho}{\partial t} = -\left(\frac{\partial}{\partial x} \rho v_1 + \frac{\partial}{\partial y} \rho v_2 + \frac{\partial}{\partial z} \rho v_3\right)$$

$$\frac{\partial \rho}{\partial t} = -\operatorname{div} \rho \mathbf{v} \quad \text{continuity eq.}$$

Prove: $\frac{d}{dt} \rho \left(\frac{v^2}{2}\right) + \operatorname{div} \rho \left(\frac{v^2}{2}\right) \mathbf{v} = \rho \frac{D}{Dt} \left(\frac{v^2}{2}\right)$

(2) $\frac{d}{dt} \rho \langle v \rangle + \operatorname{div} \rho \langle v \rangle \mathbf{v} = \rho \frac{D}{Dt} \langle v \rangle$

(1) $\frac{\partial \rho}{\partial t} \frac{v^2}{2} = -\left(\frac{\partial}{\partial x} \rho v_1 \frac{v^2}{2} + \frac{\partial}{\partial y} \rho v_2 \frac{v^2}{2} + \frac{\partial}{\partial z} \rho v_3 \frac{v^2}{2}\right)$

$$\frac{\partial \rho}{\partial t} \frac{v^2}{2} + \frac{\partial}{\partial x} \rho v_1 \frac{v^2}{2} + \frac{\partial}{\partial y} \rho v_2 \frac{v^2}{2} + \frac{\partial}{\partial z} \rho v_3 \frac{v^2}{2} = 0$$

$$\frac{\partial \rho}{\partial t} \frac{v^2}{2} + \operatorname{div} \rho \mathbf{v} \frac{v^2}{2} = 0 = \rho \frac{D}{Dt} \frac{v^2}{2}$$

(2) $\frac{\partial \rho}{\partial t} \langle v \rangle + \frac{\partial}{\partial x} \rho v_1 \langle v \rangle + \frac{\partial}{\partial y} \rho v_2 \langle v \rangle + \frac{\partial}{\partial z} \rho v_3 \langle v \rangle = 0$
 $= \rho \frac{D}{Dt} \langle v \rangle$

Using Cartesian coordinates, Prove for a fluid of constant density

$$\operatorname{div} (\nabla v)^T = 0 = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

Laplacian

$(\nabla \cdot \nabla) = 0$
 under constant density

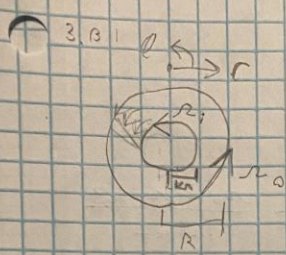
$\frac{d\rho}{dt} = \rho(\nabla \cdot \mathbf{v})$
 $\downarrow$
 const ρ

$0 = \rho(\nabla \cdot \mathbf{v})$

$\downarrow$
 $0 = (\nabla \cdot \mathbf{v})$

$\downarrow$
 $0 = \nabla \cdot (\nabla v)^T$

$$0 = \begin{bmatrix} \frac{\partial^2 v_x}{\partial x^2} & \frac{\partial^2 v_y}{\partial x^2} & \frac{\partial^2 v_z}{\partial x^2} \\ \frac{\partial^2 v_x}{\partial y^2} & \frac{\partial^2 v_y}{\partial y^2} & \frac{\partial^2 v_z}{\partial y^2} \\ \frac{\partial^2 v_x}{\partial z^2} & \frac{\partial^2 v_y}{\partial z^2} & \frac{\partial^2 v_z}{\partial z^2} \end{bmatrix}$$



APP B.4 (cylindrical)

$$\frac{\partial p}{\partial r} + \frac{1}{r} \frac{\partial}{\partial r} (p r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (p r v_\theta) + \frac{\partial}{\partial z} (p v_z) = 0$$

$$0 + 0 + \frac{1}{r} \frac{\partial}{\partial \theta} (p r v_\theta) + 0 = 0$$

- $v_z = 0 = v_r$, $v_\theta = v_\theta(r)$
 - Incompressible fluid, $T = \text{const}$ ($\rho = \text{const}$)
 - $p = p(r, z)$ $T = \text{const} = \text{in const}$
- Hydrostatic center.

APP B.8

r: $p(0 + 0 + 0 + 0 - \frac{v_\theta^2}{2}) = -\frac{\partial p}{\partial r} + \mu(0 + 0 + 0 - 0) + 0$

θ : $p(0 + 0 + 0 + 0) = 0 + \mu[\frac{\partial}{\partial r}(\frac{1}{r} \frac{\partial}{\partial r} r v_\theta) + 0 + 0 + 0] + 0$

z: $p(0) = -\frac{\partial p}{\partial z} + \mu(0) + \rho g_z$

BC1: $v_\theta|_{r=R} = \Omega_0 R_0$ BC2: $v_\theta|_{z=KR} = 0, KR$

$0 = \mu[\frac{\partial}{\partial r}(\frac{1}{r} \frac{\partial}{\partial r} r v_\theta)] \rightarrow 0 = \frac{\partial}{\partial r}(\frac{1}{r} \frac{\partial}{\partial r} r v_\theta)$

$r C_1 = \frac{\partial}{\partial r} (r v_\theta) \rightarrow \frac{r^2 C_1}{2} + C_2 = r v_\theta$

$\frac{r C_1}{2} + \frac{C_2}{r} = v_\theta$

At B.C2: $0, KR = \frac{(KR) C_1}{2} + \frac{C_2}{KR}$ $\dots = K$

At B.C1: $\Omega_0 R = \frac{R C_1}{2} + \frac{C_2}{R}$

$\Omega_0 R - \Omega_0 K^2 R = \frac{R C_1}{2} - \frac{K^2 R C_1}{2}$

$\frac{2(\Omega_0 R - \Omega_0 K^2 R)}{R} = C_1 (1 - K^2)$

$\frac{2(\Omega_0 - \Omega_0 K^2)}{(1 - K^2)} = C_1$

$$R_0 R = \frac{R}{2} \left(2(R_0 - R; k^2) \right) + \frac{C_2}{R} \quad (5)$$

$$R_0 R^2 = \frac{R^2}{2} \left(2(R_0 - R; k^2) \right) = C_2$$

$$R^2 (R_0 - R; k^2) = C_2$$

$$V_\theta = \frac{r(R_0 - R; k^2)}{1 - k^2} + \frac{R^2}{r} \left(R_0 - \frac{R_0 - R; k^2}{1 - k^2} \right)$$

$$\frac{R_0(1 - k^2)}{(1 - k^2)} - \frac{R_0 - R; k^2}{(1 - k^2)}$$

$$\frac{R^2}{r(1 - k^2)} \left(R_0 - R; k^2 - R_0 + R; k^2 \right)$$

$$= \frac{R^2 k^2}{r(1 - k^2)} \left(R_0 - R \right)^2$$

$$V_\theta = \frac{kR}{(1 - k^2)} \left[\frac{r(R_0 - R; k)}{kR} + \frac{Rk}{r} (R_0 - R) \right]$$

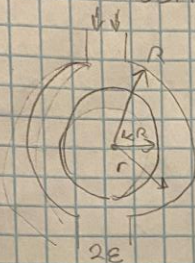
$$3 \text{ Torque} = 3 \pi \rho r^2 \omega L (kR) \quad , \quad \tau_{\theta r} = 2 \pi L (kR)$$

$$\tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{V_\theta}{r} \right) \right]$$

$$= -\mu \left[r \frac{\partial}{\partial r} \left(\frac{C_1}{2} + \frac{C_2}{r^2} \right) \right] = -\mu r \left[r \left(+3 \frac{C_2}{r^3} \right) \right]$$

$$= -3 \mu \frac{C_2}{r^2}$$

3B.4 Creeping Flow



• Fluid is very viscous

$$v_\theta = v_\theta(r, \theta) \quad v_r = 0 = v_\phi$$

• Assuming constant μ, ρ

a) APP B.4

$$\frac{\partial p}{\partial \theta} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (p v_\theta \sin \theta) + 0 = 0$$

$$\frac{1}{r \sin \theta} \cdot \frac{\partial}{\partial \theta} (p v_\theta \sin \theta) = 0$$

$$\frac{\partial}{\partial \theta} p v_\theta \sin \theta = 0 \quad \bullet p = \text{const}$$

$$p \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0$$

$$v_\theta \sin \theta = c_1 = u(r)$$

b) APP B.6

$$p \left(0 + 0 + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + 0 + 0 \right)$$

$$= -\frac{1}{r} \frac{\partial p}{\partial \theta} = \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) + \frac{1}{r^3} \frac{\partial}{\partial \theta} (r^3 \frac{\partial}{\partial \theta} (v_\theta \sin \theta)) + 0 \right]$$

$$\bullet \tau_{r\theta} = \tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + 0 \right] + p \delta_{\theta\theta}$$

$$\bullet \tau_{\theta\theta} = -\mu \left[2 \cdot \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + 0 \right] + 0$$

$$\bullet \tau_{\phi\phi} = 0$$

$$p \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \tau_{\theta\theta} \sin \theta) \right] p \delta_{\theta\theta}$$

$$= -\frac{1}{r} \frac{\partial p}{\partial \theta} + p \delta_{\theta\theta} = -\frac{1}{r} \frac{\partial p}{\partial \theta}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \tau_{\theta\theta} \sin \theta) = \frac{1}{r} \frac{\partial p}{\partial \theta}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) \cdot \sin \theta + 0 = \frac{1}{r} \frac{\partial p}{\partial \theta} \cdot \sin \theta$$

$$3B.4 \quad b) \quad \rho \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) \right)$$

$$\frac{1}{\sin \theta} \cdot \rho \frac{v_\theta}{r} \left(\frac{\partial v_\theta}{\partial \theta} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) \right)$$

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left(\frac{1}{\sin \theta} \cdot \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) \right)$$

$$c) \quad \frac{1}{r} \cdot \frac{\partial p}{\partial \theta} \cdot \sin \theta = \mu \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) \right)$$

$$\frac{\partial p}{\partial \theta} \cdot \sin \theta = \mu \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) \right) = B$$

$$BC \quad \frac{1}{2} \quad v_z|_R = 0 \quad v_z|_{\kappa a} = 0$$